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Project Summary & Project Framing

Common problem for school systems is deciding which resources to allocate which students; Not every student can receive unlimited resources from every school system. Our goal in the advising team is to determine which students will benefit the most from the allocated resources based on various demographic and academic data collected by the school system. We hope to answer the question of which students are most at risk for performing poorly on their G3 term scores based on their available demographic and academic data. To answer this we will investigate two separate models, one predicting G3 scores with the use of G1 and G2 scores and another one predicting G3 scores without the use of G1 and G2 scores.

We found that G1 and G2 scores are highly correlated to G3 scores. While we had limited success with predicting a G3 scores without using G1 and G2 scores, our model for predicting G3 scores without G1 and G2 scores could still be used to make low resource intensive decisions such as Tier 1 interventions or as a simple early warning indicator on a student's record to make faculty aware.

It is our recommendation that the linear regression model using G1 and G2 scores to predict G3 scores be used in production to determine which students should have higher tiers of intervention allocated toward them.

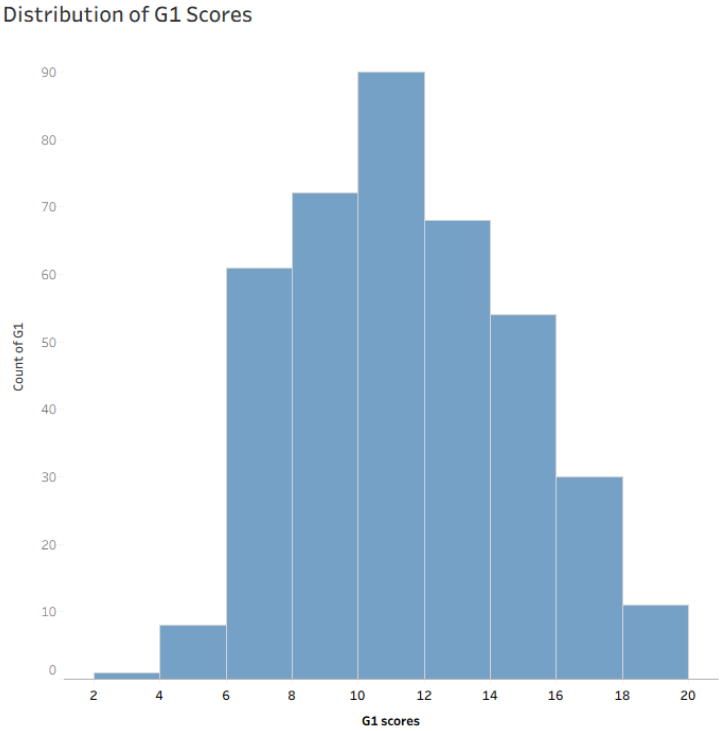

Figure 1 Distribution of G1 scores

Data Overview

The data used in this model was from UC Irvine's machine learning repository. UC Irving collected this data

from school reports and questionnaires from two Portuguese schools. It has 34 features ranging from age of the student to number of absences in a term of a student, and there are 395 rows. The data appears to be generally complete with minimal missing values. Notably, they are only missing values for the absences and the age of the recorded responses. Those missing values are 13 for absences in G1, 9 for absences and

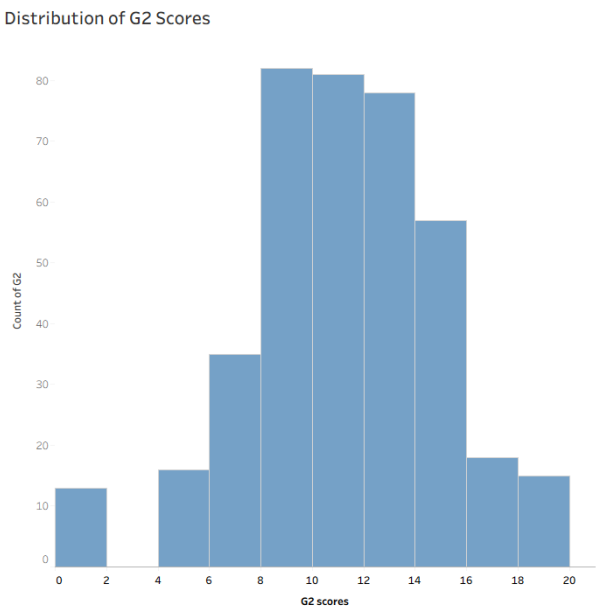

Figure 3 Distribution of G3 scores

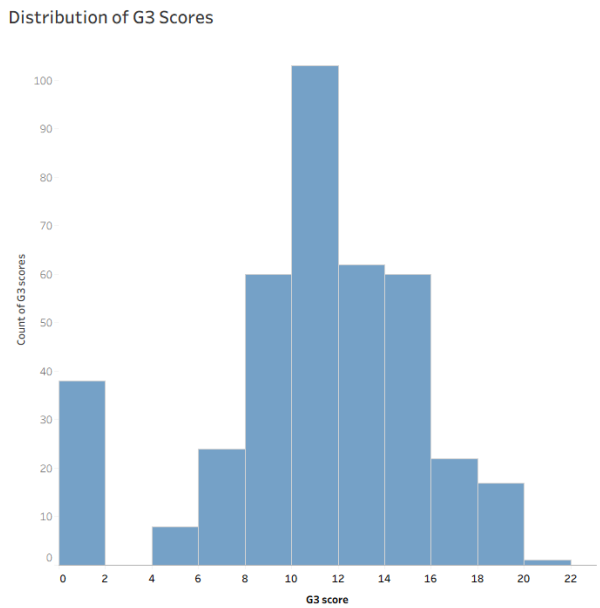

Figure 2 Distribution of G2 scores

G2, 14 for absences in G3, and 12 for the age. The features contain 16 categorical features, 8 numerical features, and 20 ordinal features.

G3 is the feature we seek to predict with our model. In Figures 1, 2, and 3 we see G1, G2, and G3 scores all appear to behave normally as we expect. We do note that in Figures 2 and 3 there do appear to be outliers in the 0-score bin. This might be due to students who have not yet received their G2 and G3 term scores. We also noticed possible negative trends when plotting age, time spent with friends, and travel time to school against G3 scores in Figures 4 and 5. There also appears to be a clear distinction in the average G3 scores plotted against a student's desire to attend higher education in Figure 6.

One problem we hope to mitigate with our model is having to wait to provide extra resources to students only after we have received their G1 and G2 scores. If we can identify whether a student is likely to have a lower G3 score based on data other than their G1 and G2 scores, we may be able to offer extra resources and intervention at earlier stages in that

student's life leading to increased G1 and G2 scores and thus, increased G3 scores. To address this, we will investigate 2 models. One model will include the G1 and G2 scores, and a separate model will exclude the G1 and G2 scores.

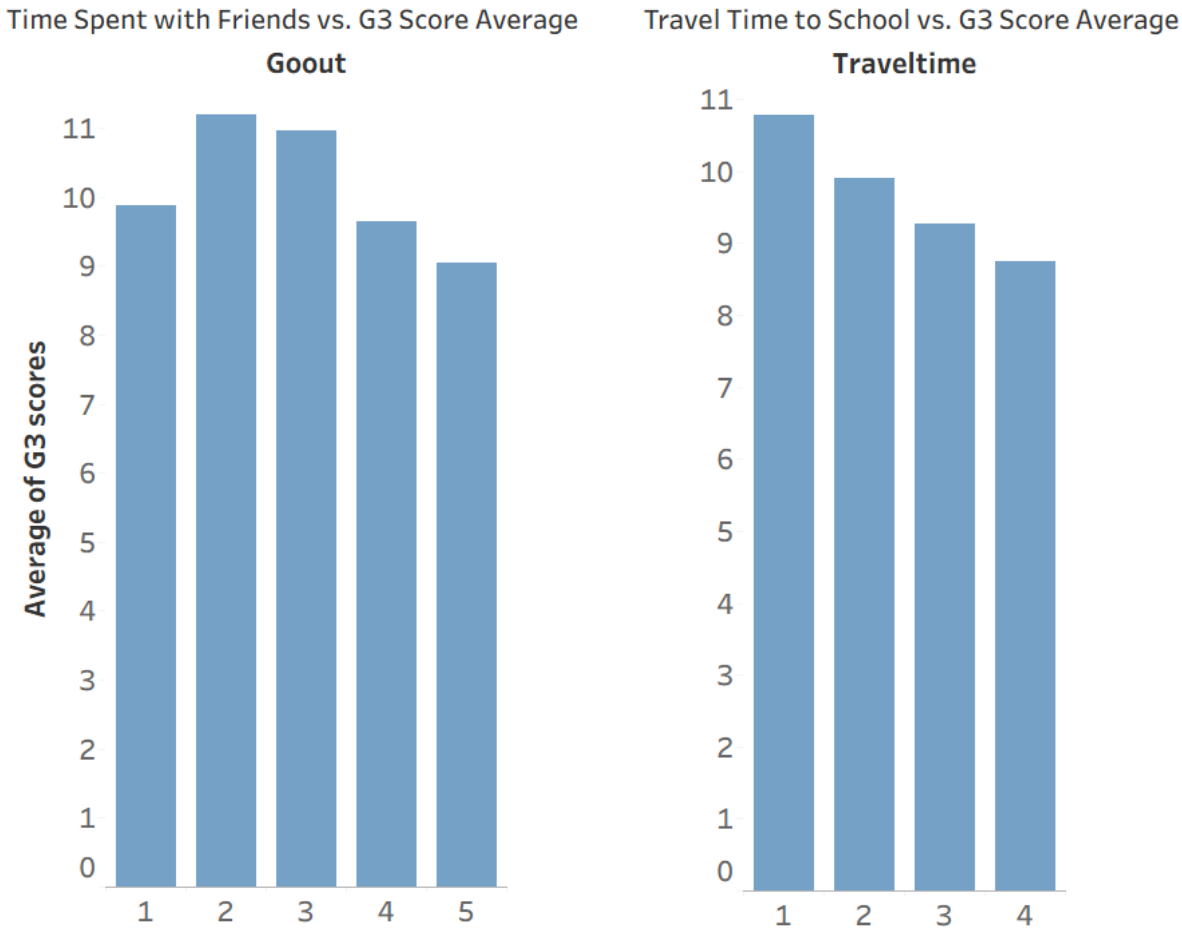

Figure 4 Time spent with friends and travel time to school versus Average G3 score

Analytical Insight

Plotting the count of G1, G2, and G3 scores against their value, we see a normal bell curve in Figures 1, 2, and 3. This is to be expected since term scores are normally distributed, and it verifies that no intensive data manipulation is needed for the G1, G2, or G3. Regarding the possible outliers, we do notice that the amount of 0-scores increased with G3 scores compared to G2 scores. This could indicate that this loss is cumulative suggesting it is students who have not received the G2 and G3 term scores yet. It is also possible that the student could have moved to a different school district before completing their G2 or G3 term.

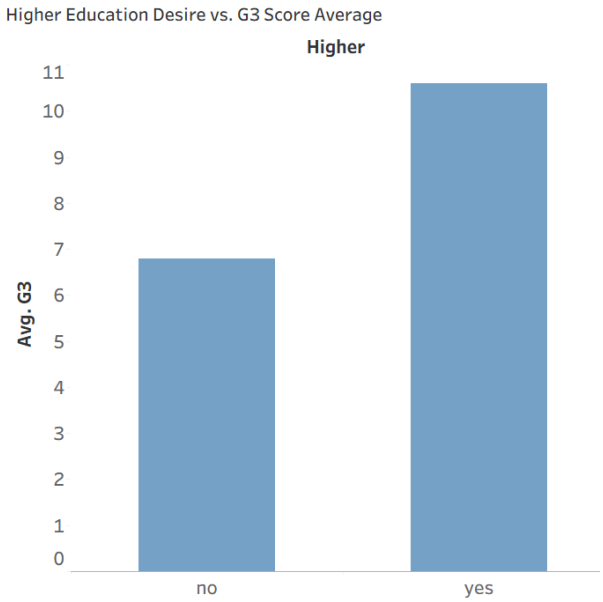

Figure 6 Desire for higher education vs average G3 scores

Figures 4 and 5 suggest possible negative relationships with the average G3 scores when plotted against age, time spent with friends, and travel time to school. Looking at the time spent with friends, it could be suggested that too little time spent with friends could lead to not enough collaboration as well as too much time spent with friends could lead to not enough studying. Both could contribute to the lower G3 score. For age versus the average G3 score in Figure 5, we notice a steady decline in scores for younger students, a dramatic spike around age 20 followed by a steep decline for older students. Students who progress in school faster along with students who are held back for low term scores could be contributing to this. The only demographic feature that showed significant difference was students who expressed an interest in pursuing higher education as seen in Figure 6.

Figure 5 Age versus G3 scores

We notice a clear trend with G1 scores, G2 scores, and the average of G1 and

G2 scores all plotted against G3 scores in Figures 7, 8, and 9. The different bubble size indicates the amount of student at that mark. This is to be expected since a student's performance on earlier exams should be indicative of their performance on later exams. This is further shown with high correlation metrics of 0.79 for G1 and 0.91 for G2.

When examining our correlation metrics for our numeric and ordinal data, we see possible correlation between the mother and father's education and term scores. The mother's correlation coefficient is around 0.21, and the father's correlation coefficient is between 0.14 to 0.19. In our model, we will attempt to only focus on demographic data solely relating to the student to account for different home lives that students might have. Not all students may have a mother and a father. The number of absences a student accrues and the G1, G2, and G3 terms also offers suspicion but not at first glance. Was very low correlation coefficients of 0.03, 0.08, and 0.11 respectively, it's not obvious that absences would contribute to G3 scores. We will instead look at the total accumulated absences which should offer a higher correlation.

The number of failures is another promising feature to track offering a -0.37 correlation coefficient. This feature is interesting because it suggests a moderate correlation, but also is easy to track in real time through the students teachers and direct support network without having to update demographic surveys of the student. These surveys could happen at irregular intervals or less than once a year which would work against the purpose of this model. In contrast a student’s teacher is easily able to report on a student’s “At-Risk” status of failing a class. For this reason we will include failures as well.

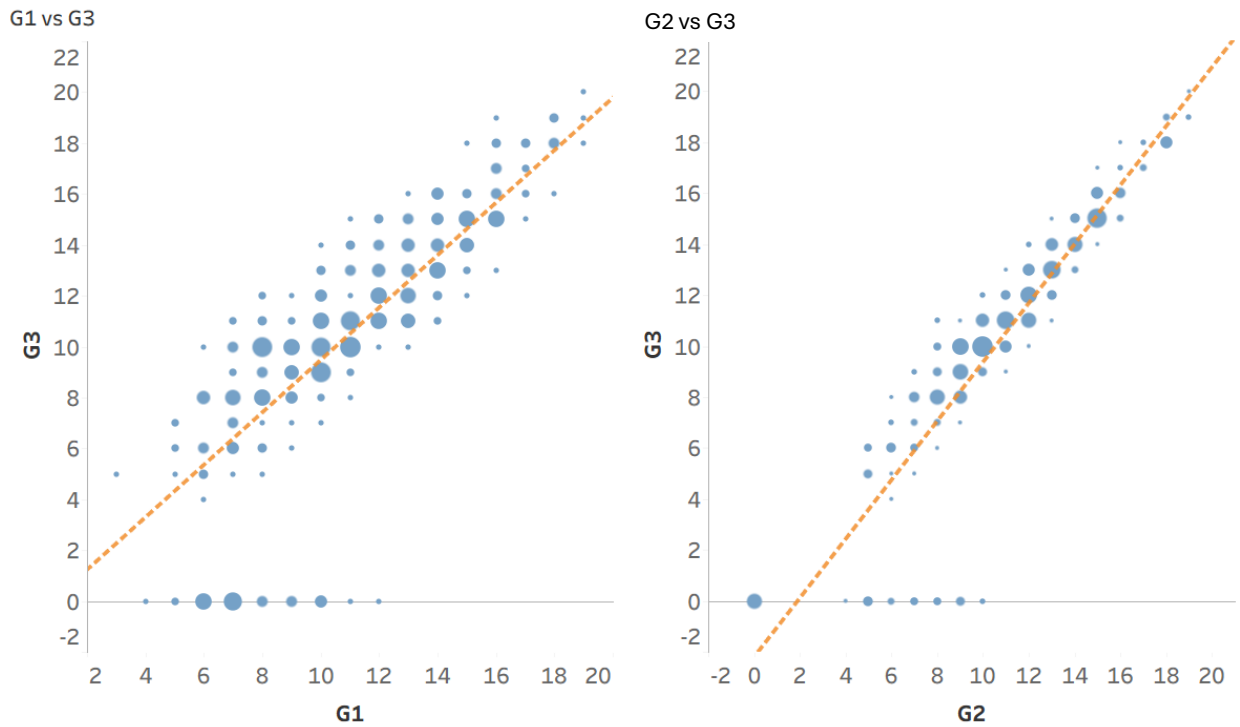

Figure 7 G1 and G2 scores vs G3 scores. The larger bubble size indicates more students are at that mark.

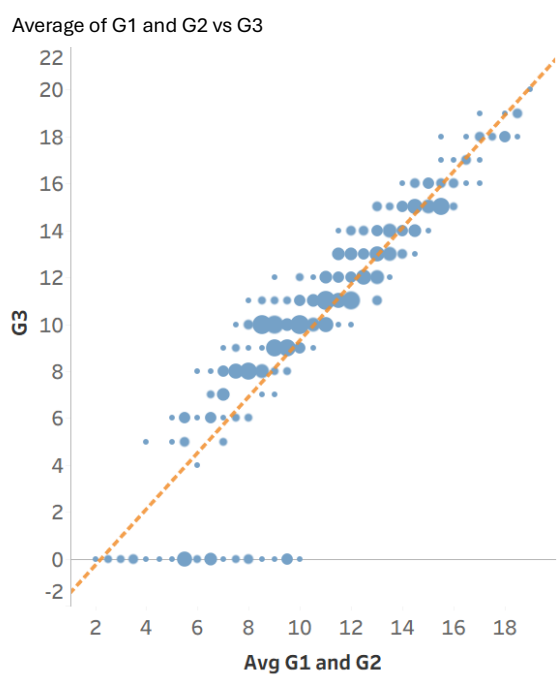

Figure 8 Combined average of G1 and G2 scores versus G3 scores

Lastly, we will include workday and weekend alcohol consumption. Although these features offer less than awesome a -0.10 correlation coefficient, a possible expansion of this model could be investigating total alcohol consumption like total absences.

Methodology

We test three models both with and without G1 and G2 scores with two metrics to evaluate their performance.

Our first model is a Linear Regression model which best tries to fit a straight line through the data while minimizing the distance all data points are to our line. We then use that line to predict new values based on the features we receive. Our second model is a Linear Support Vector Regression (LinearSVR) model. This LinearSVR attempts to divide our data using a “street.” The street divides our data into “left” and “right” sides-of-the-street sections which we can then predict what new values

will be based on our features. This is built using the data points closest to the edge of the street as "support" data points. We can change the width of this street to adjust our model as needed. Lastly, we test a Lasso Regression model. Lasso regression is like Linear Regression in that it tries to fit a straight line through the data, but it has a weighted mathematical expression that helps shrink the effects of less important features.

We evaluate these models using root mean squared error (RMSE) and R^2. RMSE increases if the predicted values are farther away from the actual values when we test our model. R^2 is between zero and one, and it is typically thought of as a percentage. A high R^2 means a large percentage of the model's variability is accounted for in the model. We attempt to keep RMSC down while keeping R^2 up.

Key Results

Model	G1/G2 scores	RMSE
Linear Regression	With	1.88
	Without	4.36
LinearSVR	With	2.09
	Without	4.40
Lasso Regression	With	2.17
	Without	4.39

Table 1 Model Metrics after tuning

After tuning our models, we achieve the lowest RMSE of 1.88 for our linear regression model including the G1 and G2 scores. (Table 1) We can see that the model performance significantly degraded whenever the G1 and G2 scores were excluded from our training data. This is further seen in the R^2 values for the LinearSVR with a value of 0.79 including the G1 and G2 scores and a value of 0.05 excluding the G1 and G2 scores.

We can recommend adding the Linear Regression model with G1 and G2 scores into production, but it should be cautioned that using the Linear Regression model without G1 and G2 scores is not advised for making major resource use decisions. A possible use of the Linear Regression model without G1 and G2 scores could be an early warning flag on the students record for low resource intensive tier 1 interventions. Tier 2 decisions and up should not be made with the Linear Regression model without G1 and G2 scores.

Examples of tier 1 interventions could be but are not limited to:

- An advisor or trusted teacher checking in with his student about failures or absences from a class
- An advisor or trusted teacher asking about plans after secondary education
- Ensuring all students are included in daily homeroom activities to prevent isolation

- An advisor or trusted teacher talking with parents about alcohol use

Conclusion

We were successful in finding a model to use to predict students G3 scores. This was due Check the pipeline management with our model, scaling our data when needed, and applying a custom transformer which increased our model's effectiveness by aggregating the total absences.

Although we were successful in finding a model to predict G3 scores, this model is limited in its capacity since it relies heavily on the G1 and G2 scores. We were unsuccessful in finding a robust model for G3 scores solely relying on demographic and social data. Some possible improvements to this model could be aggregating the total alcohol consumption of a student along with measuring the difference in study time versus time spent with friends. It seems to be that creating more calculated features from already available demographic and social data would be a significant way to improve the model.

Our scope and model diversity that we used is limited, so a future team might try using more complex models such as random forests along with increasing the sample size of the training data collected.